

connectivity between any two points on the Earth or in the air, like a balloon or a drone. However, in the Indian context, unlike Project Loon, in one of the deployment scenarios proposed, the television white-space modem technology could effectively provide last-hop wireless connectivity from the nearest optical-fibre node to rural/urban areas.

"High-speed optical connectivity is being made available at most *panchayats* by National Optical-Fibre Network (NOFN) project, BharatNet. Connectivity to high-speed optical backbone makes the television white-space modem solution quite unique to enable Digital India."

Building connectivity in Africa with digital BRCK

Bad weather, harsh terrain, poverty, name the problem and it is there in Africa. Yet, BRCK has found a way of providing connectivity to the remotest corners of this continent. And guess what their website proudly proclaims: "If it works in Africa, it will work anywhere!"

BRCK is basically a ruggedised router that can work under extremely-bad weather conditions and with very unreliable power supply.

According to the company, "It is a rugged, cloud-managed, full-featured modem/router with built-in fail-overs and programmable general-purpose input-output (GPIO) expansion. If there is a way for you to connect, BRCK will help you get up and stay up, no matter where you are."

This makes it a perfect tool for connecting rural areas. In Africa, for example, BRCK is being used extensively to provide connectivity to rural schools, so children can have access to good educational content.

All you need to do to get BRCK up and running is insert a 3G data-enabled subscriber identity module (SIM) card and the device starts broadcasting Wi-Fi signals, enabling up to 20 devices to connect to the

Internet. It works in more than 140 countries.

In case you do not have a SIM, you can opt for BRCK vMNO for global connectivity without SIM cards. There is an external GSM antenna port, which helps improve reach along the edge of a signal range. The battery, which can be charged using solar power, car battery, computer or regular power source, lasts for eight hours in full-power mode and much longer when used in low-power mode.

Plus, the robust BRCK system is designed as a platform that you can build on. You can write your own custom software for it through the application programming interface (API), connect sensors and other hardware through its Arduino-compatible GPIO ports, upgrade the 4GB onboard storage and do much more.

The most interesting part is that BRCK can be managed on the cloud. So this was, say, a non-profit organisation somewhere in New Delhi can install BRCK in a school in a Himalayan village, remotely manage its settings, push content and apps to it, among others.

Innovative devices to extend the reach of BharatNet

India is also gearing up to the connectivity challenge. We are aware that the current government is committed to take broadband connectivity to all 250,000 villages. While BharatNet, the ₹ 720 billion NOFN, will be the backbone of the project, private and government-funded companies are simultaneously developing appropriate devices to improve last-mile connectivity. Kayargadde explained one scenario using television white-space as earlier.

Wi-Fi products launched by Centre for Development of Telematics (C-DoT) in July 2015 are also good examples of this trend.

One device is a long-distance Wi-Fi system, which can solve connectivity issues in rural areas, hilly

terrain, highways, tunnels, dense forests and others.

The other is a solar-powered Wi-Fi device, which will reduce dependence on grid power. It has also developed a smart green power supply solution that can make broadband solutions self-sustainable.

These cost-effective solutions help improve the reach of BharatNet, beyond *panchayats*, to the remotest corners of the country.

Indeed, it is heartening to learn of such efforts closer home, because universal connectivity is not something we can distance ourselves from, and watch the world develop. It is a real problem faced by real people, in our own country, and requires each of us to empathise with the situation and do our bit, as engineers, entrepreneurs, philanthropists or volunteers.

While signing off, Dr Borkar leaves us thinking about how economics matters at the grassroot level. He says, "Primary issues for a product or service to customers are affordability and capabilities that can be provided at that price-point to consumers. The goal of any commercial enterprise is to make profit. For providing the Internet to customers, currently price-points are higher than what almost two-thirds of the people can afford."

"Although making universal broadband Internet available to all may be a laudable goal, economic reality dictates the imperative of providing scalable and affordable Internet with a range of prices and corresponding performance and capabilities."

"High tele-density of 2G GSM wireless system in India is a prime example of a basic voice technology being priced within the buying power of most people. Unless availability of the Internet is considered a fundamental right with significant support by the government, universal broadband Internet connectivity will stay a dream." **EFY**